MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 6: Exponential and related functions 2. Examples

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Lecture outline

1. Examples

Example 15 (Slides No 3). Part 1

Example

Show that $u(x, y) = e^{-x}(x \sin y - y \cos y)$ is harmonic.

$u_x = -e^{-x}(x \sin y - y \cos y) + e^{-x} \sin y$
$u_{xx} = e^{-x}(x \sin y - y \cos y) - e^{-x} \sin y - e^{-x} \sin y =$
$e^{-x}(x \sin y - 2 \sin y - y \cos y)$
$v_y = e^{-x}(x \cos y - \cos y + y \sin y)$
$v_{yy} = e^{-x}(-x \sin y + \sin y + \sin y + y \cos y) =$
$e^{-x}(-x \sin y + 2 \sin y + y \cos y)$
Thus $u_{xx} + v_{yy} = 0$.

Example 15 (Slides No 3). Part 2

Example

Let $u(x,y) = e^{-x}(x\sin y - y\cos y)$.
Find $v(x,y)$ such that $f(z) = u + iv$ is analytic.

$$v_y = u_x = -e^{-x}x\sin y + e^{-x}\sin y + e^{-x}y\cos y$$

$$v = \int[-e^{-x}x\sin y + e^{-x}\sin y + e^{-x}y\cos y]dy + g(x)$$

$$v = e^{-x}x\cos y - e^{-x}\cos y + e^{-x}y\sin y + e^{-x}\cos y + g(x)$$

$$v_x = e^{-x}\cos y - e^{-x}x\cos y - e^{-x}y\sin y + g'(x) =$$

$$-u_y = e^{-x}\cos y - e^{-x}x\cos y - e^{-x}y\sin y$$

$$g'(x) = 0$$

$$g(x) = C$$

$$v = e^{-x}x\cos y + e^{-x}y\sin y + C.$$

Example 1. Part 1

Example

Prove: (a) $e^{z_1}e^{z_2} = e^{z_1+z_2}$

$$e^{z_1} = e^{x_1+iy_1} = e^{x_1}[\cos y_1 + i\sin y_1]$$
$$e^{z_2} = e^{x_2+iy_2} = e^{x_2}[\cos y_2 + i\sin y_2]$$

$$e^{z_1}e^{z_2} = e^{x_1+x_2}\cos y_1\cos y_2 - e^{x_1+x_2}\sin y_1\sin y_2 +$$
$$i[e^{x_1+x_2}\cos y_1\sin y_2 + e^{x_1+x_2}\sin y_1\cos y_2]$$
$$= e^{x_1+x_2}[\cos y_1\cos y_2 - \sin y_1\sin y_2 + i[\cos y_1\sin y_2 + \sin y_1\cos y_2]]$$
$$= e^{x_1+x_2}[\cos(y_1+y_2) + i\sin(y_1+y_2)]$$
$$= e^{x_1+x_2+i(y_1+y_2)}$$
$$= e^{z_1+z_2}$$

Example 1. Part 2

Example

Prove:

(b) $|e^z| = e^x$

$$|e^z| = |e^x[\cos y + i \sin y]| = \sqrt{e^{2x}[\cos^2 y + \sin^2 y]} = e^x$$

(c) $e^{z+2\pi ni} = e^z \ (n = 0, \pm 1, \pm 2, \ldots)$

$$\begin{aligned} e^{z+2\pi ni} = e^{x+i(y+2\pi n)} &= e^x[\cos(y + 2\pi n) + i \sin(y + 2\pi n)] \\ &= e^x[\cos(y) + i \sin(y)] \qquad \text{(since } \sin y \\ \text{and } \cos y \text{ are periodic with period } 2\pi \text{ for } y \text{ real.)} \\ &= e^{x+iy} \\ &= e^z \end{aligned}$$

(d) $e^{2n\pi i} = 1.$

$$\begin{aligned} e^{2n\pi i} &= \cos 2n\pi + i \sin 2n\pi \\ &= 1 + 0i = 1. \end{aligned}$$

Example 2.

Example

Show that $\exists$ no finite value w such that $e^w = 0$.

Suppose there is a finite w such that $e^w = 0$

Then, $e^w = e^{x+iy} = e^x[\cos y + i \sin y] = 0$

Since $e^x \neq 0 \ \forall \ x$ real, then $\cos y = \sin y = 0$, which is impossible.

Therefore, there is no finite w such that $e^w = 0$.

Example 3. Part 1

Example

Establish the following

a) $\cos z = \cos x \cosh y - i \sin x \sinh y$

$$\begin{aligned}
\cos z = \frac{e^{iz} + e^{-iz}}{2} &= \frac{1}{2}[e^{ix-y} + e^{-ix+y}] \\
&= \frac{1}{2}[e^{-y}[\cos x + i \sin x] + e^{y}[\cos x - i \sin x]] \\
&= \frac{1}{2} i \sin x (e^{-y} - e^{y}) + \tfrac{1}{2} \cos x (e^{y} + e^{-y}) \\
&= -\frac{1}{2} i \sin x (e^{y} - e^{-y}) + \tfrac{1}{2} \cos x (e^{y} + e^{-y}) \\
&= -i \sin x \sinh y + \cos x \cosh y.
\end{aligned}$$

Example 3. Part 2

Example

Establish the following

b) $\sin(z+w) = \sin z \cos w + \cos z \sin w$

$$\sin(z+w) = \frac{e^{i(z+w)} - e^{-i(z+w)}}{2i}$$
$$= \frac{e^{iz}e^{iw} - e^{-iz}e^{-iw}}{2i}$$

$$= \frac{[\cos z + i \sin z][\cos w + i \sin w] - [\cos z - i \sin z][\cos w - i \sin w]}{2i}$$
$$= \frac{\cos z \cos w - \sin z \sin w - \cos z \cos w + \sin z \sin w}{2i} +$$
$$+ \frac{i \cos z \sin w + i \sin z \cos w + i \cos z \sin w + i \sin z \cos w}{2i}$$
$$= \frac{2i \cos z \sin w + 2i \sin z \cos w}{2i}$$
$$= \cos z \sin w + \sin z \cos w$$

Example 3. Part 3

Example

Establish the following

c) $\sin^2 z + \cos^2 z = 1$

$$\begin{aligned}\sin^2 z + \cos^2 z &= \left(\frac{e^{iz} - e^{-iz}}{2i}\right)^2 + \left(\frac{e^{iz} + e^{-iz}}{2}\right)^2 \\ &= \frac{e^{2iz} - 2 + e^{-2iz}}{-4} + \frac{e^{2iz} + 2 + e^{-2iz}}{4} \\ &= \tfrac{4}{4} = 1.\end{aligned}$$

Example 4. Part 1

Solve the equation $\cos z = i$

Hint: $\sinh^{-1}\beta = \ln\left(\beta + \sqrt{\beta^2+1}\right)$ for all real β.

$$\cos z = \cos x \cosh y - i \sin x \sinh y = i$$

Therefore $\cos x \cosh y = 0$.

Since $\cosh y \neq 0$, then $\cos x = 0$.

Then $x = \dfrac{(2n+1)\pi}{2}$ where n is any integer.

Also, $\sin \dfrac{(2n+1)\pi}{2} \sinh y = (-1)^n \sinh y = -1$.

Thus $y = \begin{cases} \sinh^{-1}(-1) & n \text{ even} \\ \sinh^{-1}(+1) & n \text{ odd} \end{cases}$

Example 4. Part 2

Hence the solutions of $\cos z = i$ are

$$\frac{2n+1}{2}\pi + i\sinh^{-1}(-1), \qquad n \text{ even}$$
$$\frac{2n+1}{2}\pi + i\sinh^{-1}(1), \qquad n \text{ odd}$$

But $\sinh^{-1}\beta = \ln\left(\beta + \sqrt{\beta^2+1}\right)$ for all real β, so

$$\begin{aligned} \sinh^{-1}(-1) &= \ln\left(-1+\sqrt{2}\right) \\ \sinh^{-1}(1) &= \ln\left(1+\sqrt{2}\right) \end{aligned}$$

Therefore

$$\begin{aligned} z &= \frac{2n+1}{2}\pi + i\ln\left(-1+\sqrt{2}\right), \qquad n \text{ even} \\ z &= \frac{2n+1}{2}\pi + i\ln\left(1+\sqrt{2}\right), \qquad n \text{ odd} \end{aligned}$$

Example 5

Example

Prove that $\sin(\frac{\pi}{2} - z) = \cos z$

$$\begin{aligned}\sin(\tfrac{\pi}{2} - z) &= \sin(\tfrac{\pi}{2})\cos(-z) + \cos\left(\tfrac{\pi}{2}\right)\sin(-z) \\ &= \cos(z) + 0 = \cos z.\end{aligned}$$

Example 6. Part 1

Example

a) Find $\log\left[i^{\frac{1}{4}}\right]$

$$\begin{aligned}\log[i^{\frac{1}{4}}] &= \log[\cos(\tfrac{\pi}{8}) + i\sin(\tfrac{\pi}{8})] \\ &= \ln(1) + (\tfrac{\pi}{8} + 2n\pi)i \qquad (n = 0, \pm 1, \pm 2, ...) \\ &= (\tfrac{\pi}{8} + 2n\pi)i\end{aligned}$$

Also, $i^{\frac{1}{4}} = (\cos(\frac{3\pi}{8}) + i\sin(\frac{3\pi}{8})), (\cos(\frac{5\pi}{8}) + i\sin(\frac{5\pi}{8})), (\cos(\frac{7\pi}{8}) + i\sin(\frac{7\pi}{8}))$

So, $\log[\cos(\frac{3\pi}{8}) + i\sin(\frac{3\pi}{8})]$

$$\begin{aligned}&= \ln(1) + (\tfrac{3\pi}{8} + 2n\pi)i \qquad (n = 0, \pm 1, \pm 2, ...) \\ &= (\tfrac{3\pi}{8} + 2n\pi)i\end{aligned}$$

$\log[\cos(\frac{5\pi}{8}) + i\sin(\frac{5\pi}{8})]$

$$\begin{aligned}&= \ln(1) + (\tfrac{5\pi}{8} + 2n\pi)i \qquad (n = 0, \pm 1, \pm 2, ...) \\ &= (\tfrac{5\pi}{8} + 2n\pi)i\end{aligned}$$

$\log[\cos(\frac{7\pi}{8}) + i\sin(\frac{7\pi}{8})]$

$$\begin{aligned}&= \ln(1) + (\tfrac{7\pi}{8} + 2n\pi)i \qquad (n = 0, \pm 1, \pm 2, ...) \\ &= (\tfrac{7\pi}{8} + 2n\pi)i\end{aligned}$$

Example 6. Part 2

Example

b) Find the principal value of $\log(1+\sqrt{3}i)$.

$$\left|1+\sqrt{3}i\right| = \sqrt{1+3} = 2$$

Since $1+\sqrt{3}i$ is in the first quadrant,

$$\arg(1+\sqrt{3}i) = \frac{\pi}{3} + 2n\pi \qquad (n = 0, \pm 1, \pm 2, ...)$$

The principal value of the argument is $\frac{\pi}{3}$.

So, $\quad \log(1+\sqrt{3}i) = \ln(2) + \frac{\pi}{3}i$

Example 7

Example

Compute all values of 2^i.

$$2^i = e^{i\log(2)}$$

$$\log(2) = \ln(2) + 2n\pi i \qquad (n = 0, \pm 1, \pm 2, ...)$$

$$\text{So,} \quad e^{i\log(2)} = e^{i[\ln(2)+2n\pi i]}$$

$$= e^{-2n\pi + \ln(2)i}$$

$$= e^{-2n\pi}[\cos(\ln(2)) + i\sin(\ln(2)). \qquad (n = 0, \pm 1, \pm 2, ...)$$

Example 8

Example

Calculate all values of $(1-i)^{1+i}$

$$(1-i)^{1+i} = e^{(1+i)\log(1-i)}$$
$$|1-i| = \sqrt{1+1} = \sqrt{2}$$
$$\arg(1-i) = -\tfrac{\pi}{4} + 2n\pi \qquad (n = 0, \pm 1, \pm 2, ...)$$
$$\text{So,} \quad \log(1-i) = \tfrac{1}{2}\ln(2) + (-\tfrac{\pi}{4} + 2n\pi)i$$
$$\text{So,} \quad (1-i)^{1+i} = e^{(1+i)(\frac{1}{2}\ln(2)+(-\frac{\pi}{4}+2n\pi)i)}$$
$$= e^{\frac{1}{2}\ln(2)+\frac{\pi}{4}-2n\pi+(\frac{1}{2}\ln(2)-\frac{\pi}{4}+2n\pi)i}$$

$$= \sqrt{2}e^{\frac{\pi}{4}-2n\pi}\left(\cos\left(\ln\sqrt{2}-\tfrac{\pi}{4}\right) + i\sin\left(\ln\sqrt{2}-\tfrac{\pi}{4}\right)\right)$$

Thank you